Data Mining Fundamentals: Final Exam Study Guide

This study guide is designed to help you prepare for a 40-question multiple-choice final exam by covering the core concepts of the data mining pipeline, data warehousing, and pre-processing techniques as detailed in the sources.

Section 1: Essential Definitions and Concepts

Term	Definition (Based on Sources)
Data Mining	Also known as **Knowledge Discovery from Data (KDD)**; the process of finding interesting patterns within massive amounts of data.
Interesting Pattern	A pattern that is **new** (previously unknown), **valid** (correct), **potentially useful**, and **understandable by humans**.
Data Warehouse	A **subject-oriented**, **integrated**, **time-variant**, and **non-volatile** collection of data used to support management decision-making.
OLTP	**Online Transactional Processing**; supports daily operations and detailed, regular updates like bank transactions or course registrations.
OLAP	**Online Analytical Processing**; focuses on data analysis and identifying new patterns to support decision-making.
Interquartile Range (IQR)	The difference between the third quartile (**Q3**) and the first quartile (**Q1**).
Correlation	A measure of relationship between attributes; a key principle is that **correlation does not imply causality**.
Normalization	The process of scaling attribute values (e.g., Min-Max or Z-score) so they are directly comparable.
Discretization	Converting continuous numerical values into discrete intervals or "buckets".
Sampling	Selecting a subset of data points that is representative of the original dataset to reduce size.

Section 2: Multiple Choice Practice Questions

Topic 1: Data Mining Fundamentals & The Pipeline

1. Data mining is most accurately described as which of the following?

A. Storing data in a secure cloud environment
B. Knowledge Discovery from Data (KDD)
C. Building faster computer hardware
D. Manually entering records into a database

Answer: B. Knowledge Discovery from Data (KDD)

2. For a pattern to be considered "interesting" in data mining, it must be:

A. Complex and difficult to calculate
B. Previously known to the users
C. Potentially useful and understandable by humans
D. Applicable only to a very small subset of data

Answer: C. Potentially useful and understandable by humans

3. Which of the "Four Views" focuses on the specific usage scenario and domain knowledge?

A. Data View
B. Application View
C. Technique View
D. Knowledge View

Answer: B. Application View

4. We are currently estimated to be dealing with global data on the order of:

A. Gigabytes
B. Terabytes
C. Petabytes
D. Tens of Zettabytes

Answer: D. Tens of Zettabytes

5. What is the primary motivation for automating massive data analysis?

A. To eliminate the need for data scientists
B. To reduce the cost of cloud storage
C. Because we are "drowning in data but starving for knowledge"
D. To increase the speed of internet transmission

Answer: C. Because we are "drowning in data but starving for knowledge"

Topic 2: The "Vs" of Big Data

6. Which "V" refers to the quality and trustworthiness of the data?

A. Volume

B. Velocity

C. Veracity

D. Variety

Answer: C. Veracity

7. The "V" that describes the speed at which data is generated and must be analyzed is:

A. Volume

B. Velocity

C. Variety

D. Value

Answer: B. Velocity

8. Which "V" relates to the ultimate goal of bringing out useful insights for real-world settings?

A. Veracity

B. Variety

C. Value

D. Volume

Answer: C. Value

Topic 3: Data Understanding & Statistics

9. Which of the following is used to calculate the Interquartile Range (IQR)?

A. Max - Min

B. Mean / Standard Deviation

C. Q3 - Q1

D. (Q1 + Q3) / 2

Answer: C. Q3 - Q1

10. A scatter plot of points moving from the bottom-left to the top-right indicates:

A. Negative correlation

B. No correlation

C. Positive correlation

D. Causality

Answer: C. Positive correlation

11. Which statistical measure represents the "central tendency" by finding the middle value?

A. Mean

B. Median

C. Standard Deviation

D. Variance

Answer: B. Median

12. In the "pipeline" project example, which visualization was noted as helpful for showing distributions and identifying outliers?

A. Pie Chart

B. Boxplot

C. Line Graph

D. Heatmap

Answer: B. Boxplot

Topic 4: Data Pre-processing (Cleaning & Integration)

13. Which step involves addressing missing values, noise, and inconsistencies?

A. Data Selection

B. Data Cleaning

C. Data Modeling

D. Data Warehouse Design

Answer: B. Data Cleaning

14. What is a "noisy" data point most likely to be?

A. A missing value

B. A duplicate record

C. An error or an outlier

D. A categorical label

Answer: C. An error or an outlier

15. Using the attribute mean to fill in a missing value is an example of:

A. Manual filling

B. Automated filling

C. Data reduction

D. Data discretization

Answer: B. Automated filling

16. Entity Identification is a key task during which pre-processing phase?

A. Data Cleaning
B. Data Integration
C. Data Transformation
D. Data Selection

Answer: B. Data Integration

17. If a data mining task uses only relevant data for a specific decision-making scenario, it is said to be:

A. Integrated
B. Subject-oriented
C. Time-variant
D. Non-volatile

Answer: B. Subject-oriented

Topic 5: Similarity & Distance Measures

18. Which distance measure is calculated as the sum of absolute differences across dimensions?

A. Euclidean distance
B. Manhattan (City Block) distance
C. Minkowski distance
D. Cosine similarity

Answer: B. Manhattan (City Block) distance

19. Cosine similarity is particularly effective for:

A. Low-dimensional dense data
B. Sparse, high-dimensional text documents
C. Binary symmetric attributes
D. Calculating the mean of a dataset

Answer: B. Sparse, high-dimensional text documents

20. When comparing two objects with asymmetric binary attributes, which coefficient is often used to ignore "null" (0-0) matches?

A. Euclidean coefficient
B. Pearson coefficient
C. Jaccard coefficient
D. Manhattan coefficient

Answer: C. Jaccard coefficient

21. The "Curse of Dimensionality" refers to:

A. Not having enough attributes for a model
B. The difficulty of distance functions in differentiating neighbors in high-dimensional sparse spaces
C. The high cost of 3D visualization tools
D. Having too many records in a single CSV file

Answer: B. The difficulty of distance functions in differentiating neighbors in high-dimensional sparse spaces

Topic 6: Data Transformation & Reduction

22. Min-Max normalization scales data to a new range, typically:

A. [-1, 1]
B. [0, 1]
C. [Mean, Max]
D. [Minimum, Median]

Answer: B. [0, 1]

23. Which normalization technique utilizes the mean and standard deviation?

A. Min-Max normalization
B. Z-score normalization (Standardization)
C. Decimal scaling
D. Logarithmic transformation

Answer: B. Z-score normalization (Standardization)

24. Reducing the number of attributes (columns) in a dataset is called:

A. Numerosity reduction
B. Dimensionality reduction
C. Data discretization
D. Data aggregation

Answer: B. Dimensionality reduction

25. In "Forward Selection," the process starts with:

A. All attributes and removes the least informative one
B. Zero attributes and adds the most informative one each step
C. A random sample of attributes
D. Only the class labels

Answer: B. Zero attributes and adds the most informative one each step

26. Which sampling method ensures that each class is represented proportionately in the sample?

A. Random sampling with replacement
B. Random sampling without replacement
C. Stratified sampling
D. Cluster sampling

Answer: C. Stratified sampling

Topic 7: Data Warehousing & Cubes

27. A data warehouse characteristic that means data is appended rather than updated "in place" is:

A. Integrated
B. Time-variant
C. Non-volatile
D. Subject-oriented

Answer: C. Non-volatile

28. Which schema consists of one central fact table and multiple dimension tables without further normalization?

A. Star Schema
B. Snowflake Schema
C. Fact Constellation
D. Galaxy Schema

Answer: A. Star Schema

29. Moving from "Day" to "Month" in a data cube is an operation known as:

A. Drill-down
B. Roll-up
C. Slice
D. Dice

Answer: B. Roll-up

30. A "Dice" operation in a data cube results in:

A. A single slice of the cube
B. A sub-cube covering multiple dimensions
C. A rotated view of the data
D. Aggregated totals for the whole cube

Answer: B. A sub-cube covering multiple dimensions

31. "Full Materialization" of a data cube means:

A. Storing no pre-computed values
B. Pre-computing and storing all possible cuboids
C. Calculating values only on demand
D. Deleting the base raw data

Answer: B. Pre-computing and storing all possible cuboids

32. What is an "Iceberg Cube"?

A. A cube stored in cold storage
B. A cube that only computes cells meeting a minimum threshold (e.g., Sales > 10k)
C. A cube used exclusively for environmental data
D. A cube that has been fully materialized

Answer: B. A cube that only computes cells meeting a minimum threshold (e.g., Sales > 10k)

Topic 8: Data Mining Tasks & Techniques

33. Which task is "unsupervised" because it does not use predefined class labels?

A. Classification
B. Clustering
C. Numerical Prediction
D. Regression

Answer: B. Clustering

34. Finding items that frequently occur together (like Milk and Bread) is the goal of:

A. Frequent Pattern Analysis
B. Anomaly Detection
C. Normalization
D. Data Integration

Answer: A. Frequent Pattern Analysis

35. The primary purpose of Anomaly Detection is to find:

A. The most common patterns
B. Patterns that differ significantly from the norm
C. The average value of all objects
D. Missing values in a dataset

Answer: B. Patterns that differ significantly from the norm

Section 3: Practical Calculations & Homework Concepts

To succeed in the final, you should be able to perform these operations as seen in the homework assignments:

- **Min-Max Normalization** 15 16 **:** Formula: $v' = \frac{v-min}{max-min} \times (new\_max - new\_min) + new\_min$
- **Z-Score Normalization** 15 17 **:** Formula: $v' = \frac{v-mean}{std\_dev}$
- **Aggregation** 18 ··· **:** Grouping data by a specific attribute (e.g., `year` or `category`) and summing a measure (e.g., `sales`).
- **Similarity of Nominal Attributes** 21 **:** If two values are the same, $similarity = 1$ ($distance = 0$). If they are different, $similarity = 0$ ($distance = 1$).
- **IQR Calculation** 6 **:** Sort your data, find the median, find the medians of the two halves (Q1 and Q3), and subtract.

Section 4: The Data Mining Pipeline Quick Review

1. Data Understanding: Identify attribute types and basic statistics.

2. Pre-processing: Clean the "messy" real-world data (fix missing/noisy values).

3. Data Warehousing: Move data into a multi-dimensional model (Cubes) for OLAP.

4. Modeling: Apply algorithms (Classification, Clustering, etc.) to find patterns.

5. Evaluation: Use metrics (Accuracy, Error Rate) to determine if patterns are truly "interesting".

To help you reach your goal of 80 practice questions, here is a second set of 45 additional multiple-choice questions based on the technical details, coding assignments, and lecture transcripts in the sources.

Section 1: Data Cube Implementation & Python Basics

1. In the Pandas library, which function is used to combine data by a specific attribute (like "year") before calculating a sum?

A. .combine()

B. .groupby()

C. .aggregate()

D. .sort_values()

Answer: B. .groupby()

2. When creating a 3D scatter plot of a data cube, why might you need to use pd.Categorical(y).codes?

A. To hide sensitive data names

B. To reduce the file size of the plot

C. To convert text categories (like "bikes") into numeric values for the axis

D. To sort the data alphabetically

Answer: C. To convert text categories (like "bikes") into numeric values for the axis

3. Which Pandas method is used to turn a GroupBy object back into a standard DataFrame where the grouping attributes are columns again?

A. .to_frame()

B. .reset_index()

C. .flatten()

D. .de_index()

Answer: B. .reset_index()

4. If you are aggregating sales by "Country" and "Category" into a Python dictionary, what would the keys most likely be?

A. A list of all sales numbers

B. The total sum of all sales

C. A tuple or index representing the Country and Category

D. The filename of the dataset

Answer: C. A tuple or index representing the Country and Category

5. Which visualization tool was used in the homework to create a 3D projection of aggregated sales data?

A. Seaborn

B. Matplotlib (specifically Axes3D)

C. Tableau

D. Excel

Answer: B. Matplotlib (specifically Axes3D)

Section 2: Data Mining Fundamentals & Big Data Statistics

6. According to the sources, the Rubin Observatory generates how much data per night?

A. 20 Gigabytes

B. 20 Megabytes

C. 20 Terabytes

D. 20 Petabytes

Answer: C. 20 Terabytes

7. Approximately how many internet users were cited as being part of the modern digital era?

A. 1 Billion

B. More than 4 Billion

C. 10 Billion

D. 500 Million

Answer: B. More than 4 Billion

8. Which view of data mining focuses on the specific usage scenario, such as healthcare or business intelligence?

A. Data View
B. Technique View
C. Application View
D. Knowledge View

Answer: C. Application View

9. The goal of "Prescriptive Analytics" is to:

A. Describe what happened in the past
B. Predict what will happen in the future
C. Recommend a specific strategy or solution
D. Organize data into a warehouse

Answer: C. Recommend a specific strategy or solution

10. A "black box" model is considered less "interesting" because it lacks:

A. Accuracy
B. Understandability by humans
C. Validity
D. Speed

Answer: B. Understandability by humans

Section 3: Data Quality & Pre-processing Issues

11. Which data quality metric at the "dataset level" refers to whether metadata exists to explain what columns mean?

A. Accessibility
B. Interpretability
C. Timeliness
D. Relevance

Answer: B. Interpretability

12. The difference between "Precision" and "Accuracy" in sensor data is that Precision refers to:

A. How correct the value is
B. How fine-grained or sensitive the readings are
C. How quickly the reading is taken
D. How relevant the reading is

Answer: B. How fine-grained or sensitive the readings are

13. "Granularity" in a spatial-temporal dataset refers to:

A. The number of missing values
B. The trust you have in the provider
C. The resolution of the data (e.g., city-level vs. meter-level)
D. The color of the data points

Answer: C. The resolution of the data (e.g., city-level vs. meter-level)

14. Seeing an "Age" value of -10 in a dataset is an example of:

A. Incomplete data
B. Noisy data (an error)
C. Inconsistent data
D. Redundant data

Answer: B. Noisy data (an error)

15. If a student's "Birth Date" and "Age" do not match, the data is considered:

A. Incomplete
B. Noisy
C. Inconsistent
D. Scaled

Answer: C. Inconsistent

Section 4: Missing Values & Noisy Data Handling

16. Which method of filling missing values is considered the most "scalable" for large datasets?

A. Manual entry by a human
B. Automated methods (like attribute mean)
C. Deleting the entire database
D. Ignoring the problem

Answer: B. Automated methods (like attribute mean)

17. "Smoothing" is a technique specifically used to address:

A. Missing values
B. Noisy data
C. Dimensionality reduction
D. Data integration

Answer: B. Noisy data

18. In "Class Mean" imputation, you fill a missing value based on:

A. The average of every object in the column
B. The average of objects that belong to the same category/class
C. A random number between 0 and 1
D. The maximum value in the column

Answer: B. The average of objects that belong to the same category/class

19. Which advanced technique uses similar neighboring objects to estimate a missing value?

A. Linear Regression
B. K-Nearest Neighbor (KNN)
C. Binning
D. Normalization

Answer: B. K-Nearest Neighbor (KNN)

20. "Outliers" in a clustering task are often identified as:

A. Points in the center of a dense group
B. Points that do not fit well into any cluster
C. Points with missing values
D. Points that have been normalized

Answer: B. Points that do not fit well into any cluster

Section 5: Correlation & Causality

21. In the "Ice cream sales vs. Drowning" example, what is the key takeaway?

A. Ice cream causes cramps that lead to drowning
B. Drowning makes people want to buy ice cream
C. Correlation does not imply causality
D. You should never eat ice cream at the beach

Answer: C. Correlation does not imply causality

22. The Chi-Square test for nominal attributes compares Observed values to:

A. The Mean
B. Expected values (assuming independence)
C. The Standard Deviation
D. The Median

Answer: B. Expected values (assuming independence)

23. If a Chi-Square test shows a very large difference between observed and expected values, it suggests:

A. The attributes are independent
B. The attributes are likely correlated
C. The data is missing
D. The data is noisy

Answer: B. The attributes are likely correlated

24. A correlation coefficient of -1 indicates:

A. No relationship
B. A perfect positive correlation
C. A perfect negative correlation
D. A missing value

Answer: C. A perfect negative correlation

25. Why is it dangerous to infer causality from correlation in data mining?

A. It makes the algorithms run slower
B. It increases the volume of data
C. It can lead to incorrect real-world actions based on coincidence
D. It deletes the raw data

Answer: C. It can lead to incorrect real-world actions based on coincidence

Section 6: Normalization & Discretization

26. Transforming a city name into a "State" name is an example of:

A. Aggregation
B. Generalization
C. Normalization
D. Attribute Construction

Answer: B. Generalization

27. If you create a new attribute "Area" by multiplying "Length" and "Width", you are performing:

A. Discretization
B. Attribute Construction
C. Generalization
D. Smoothing

Answer: B. Attribute Construction

28. "Equal-Width" binning for the age group 0-100 with 5 bins would result in intervals of:

A. 20 years each (0-20, 21-40, etc.)
B. Varies based on how many people are in each group
C. 5 years each
D. 100 years each

Answer: A. 20 years each (0-20, 21-40, etc.)

29. "Equal-Frequency" binning is determined by:

A. The range of the values
B. The distribution/count of data points
C. The mean of the data
D. The number of categories

Answer: B. The distribution/count of data points

30. Supervised discretization uses what to guide the creation of intervals?

A. Predefined class labels
B. Random guessing
C. The Manhattan distance
D. The time the data was collected

Answer: A. Predefined class labels

Section 7: Data Reduction Techniques

31. Which reduction method reduces the number of rows (objects) rather than columns?

A. Dimensionality Reduction
B. Numerosity Reduction
C. Attribute Selection
D. PCA

Answer: B. Numerosity Reduction

32. "Backward Elimination" in attribute selection starts with:

A. Zero attributes
B. The full set of attributes
C. A random sample
D. Only one attribute

Answer: B. The full set of attributes

33. PCA stands for:

A. Private Component Analysis
B. Principal Component Analysis
C. Primary Category Aggregation
D. Pattern Correlation Assessment

Answer: B. Principal Component Analysis

34. Which technique is commonly used in image processing to reduce data by keeping only significant "coefficients"?

A. Linear Regression
B. Wavelet Transformation
C. Binning
D. Stratified Sampling

Answer: B. Wavelet Transformation

35. A "Parametric" reduction method like Linear Regression allows you to:

A. Store only a sample of the data
B. Store only the model parameters (slope/offset) instead of raw data
C. Increase the number of dimensions
D. Ignore the mean of the data

Answer: B. Store only the model parameters (slope/offset) instead of raw data

Section 8: Data Warehouse Schemas & Operations

36. Which schema is an extension of the Star Schema where dimension tables are normalized?

A. Galaxy Schema
B. Snowflake Schema
C. Fact Constellation
D. Planet Schema

Answer: B. Snowflake Schema

37. A Fact Constellation schema is characterized by having:

A. No fact tables
B. Multiple fact tables sharing dimension tables
C. Only one dimension table
D. A circular design

Answer: B. Multiple fact tables sharing dimension tables

38. The "Drill-down" operation in a data cube provides:

A. More aggregated data
B. Finer grained, more detailed data
C. A different 2D view of the cube
D. A sub-set of the cube

Answer: B. Finer grained, more detailed data

39. Which cube operation is used to rotate the axes of a view to look at different dimensions?

A. Slice
B. Pivot
C. Roll-up
D. Dice

Answer: B. Pivot

40. If you select only "March 2024" from a time dimension, you are performing a:

A. Slice
B. Dice
C. Drill-down
D. Pivot

Answer: A. Slice

Section 9: Materialization, Architecture & Ethics

41. "None-Materialization" (On-demand) has the primary disadvantage of:

A. Using too much disk space
B. High latency/waiting time for the user
C. Computing unnecessary data
D. Requiring manual data entry

Answer: B. High latency/waiting time for the user

42. In the "Snowflake" cloud data warehouse architecture, which three layers are separated for scalability?

A. Data, Knowledge, and Application
B. Storage, Compute, and Service
C. Cleaning, Integration, and Reduction
D. User, Server, and Internet

Answer: B. Storage, Compute, and Service

43. Which ETL step involves the pre-processing tasks like cleaning and normalization?

A. Extract
B. Transform
C. Load
D. Evaluate

Answer: B. Transform

44. "Data Ownership" and "Bias" are key concerns within the field of:

A. Data Warehousing
B. Data Ethics
C. Similarity Measures
D. Histogram Construction

Answer: B. Data Ethics

45. Anomaly detection is often used in security to find:

A. Frequent patterns
B. Suspicious or fraudulent activities
C. The average user's behavior
D. More data sources

Answer: B. Suspicious or fraudulent activities